

Complex Analysis Exam Problems

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Fall 2025

Part I

1. Define $\gamma: [0, 2\pi] \rightarrow \mathbb{C}$ by

$$\gamma(t) = e^{it}.$$

Compute

$$\int_{\gamma} z^n dz$$

for every integer n .

Solution. Since

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi,$$

we have

$$\gamma'(t) = ie^{it}.$$

Therefore,

$$\int_{\gamma} z^n dz = \int_0^{2\pi} \gamma(t)^n \gamma'(t) dt = i \int_0^{2\pi} e^{i(n+1)t} dt.$$

If $n \neq -1$, then

$$\begin{aligned} \int_{\gamma} z^n dz &= i \left[\frac{e^{i(n+1)t}}{i(n+1)} \right]_0^{2\pi} \\ &= \frac{e^{2\pi i(n+1)} - 1}{n+1}. \end{aligned}$$

Because $n+1$ is an integer,

$$e^{2\pi i(n+1)} = 1,$$

and hence

$$\int_{\gamma} z^n dz = 0.$$

If $n = -1$, then

$$\int_{\gamma} z^{-1} dz = i \int_0^{2\pi} 1 dt = 2\pi i.$$

Thus

$$\int_{\gamma} z^n dz = \begin{cases} 2\pi i, & n = -1, \\ 0, & n \neq -1. \end{cases}$$

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2. Let γ be the closed polygon

$$[1 - i, 1 + i, -1 + i, -1 - i, 1 - i].$$

Find

$$\int_{\gamma} \frac{1}{z} dz.$$

3. 1. Show that a Möbius transformation has 0 and ∞ as its only fixed points if and only if it is a dilation, but not the identity.
2. Show that a Möbius transformation has ∞ as its only fixed point if and only if it is a translation, but not the identity.

4. Let G be a region, and suppose that $f: G \rightarrow \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

5. Let

$$f(z) = e^x \sin y + ie^x \cos y, \quad z = x + iy.$$

Determine whether f is analytic on the complex plane.

6. Map

$$G = \mathbb{C} \setminus \{z : -9 \leq z \leq 9\}$$

onto the open unit disk by an analytic function f . Can f be one-to-one?

7. Give the principal branch of

$$\sqrt{1 - z^3}.$$

Solution. The principal branch of the square root is defined by

$$\sqrt{w} = \exp\left(\frac{1}{2}w\right), \quad w \in \mathbb{C} \setminus (-\infty, 0],$$

where $\log$ denotes the principal logarithm.

Therefore, the principal branch of $\sqrt{1 - z^3}$ is

$$\sqrt{1 - z^3} = \exp\left(\frac{1}{2}(1 - z^3)\right).$$

Its domain consists of those z for which

$$1 - z^3 \notin (-\infty, 0].$$

Equivalently,

$$z^3 \notin [1, \infty).$$

Writing $z = re^{i\theta}$, the condition $z^3 \in [1, \infty)$ holds precisely when

$$r \geq 1 \quad \text{and} \quad \theta = \frac{2\pi k}{3}, \quad k = 0, 1, 2.$$

Thus the domain of the principal branch is

$$\mathbb{C} \setminus \bigcup_{k=0}^2 \{re^{2\pi ik/3} : r \geq 1\}.$$

Hence the three branch cuts are the rays beginning at the cube roots of unity $1, e^{2\pi i/3}, e^{4\pi i/3}$ and extending outward. This branch is characterized by

$$\sqrt{1 - 0^3} = 1.$$

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Part II

8. Compute the integral

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz,$$

where γ is the closed path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi.$$

Solution. The path

$$\gamma(t) = 3e^{it}, \quad 0 \leq t \leq 2\pi,$$

is the positively oriented circle $|z| = 3$. Of the two singularities $z = 2$ and $z = 5$, only $z = 2$ lies inside γ .

Set

$$h(z) = \frac{e^z}{(z-5)^5}.$$

Since $z = 5$ lies outside $|z| = 3$, the function h is analytic on and inside γ . Therefore, by Cauchy's integral formula for derivatives,

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz = \int_{\gamma} \frac{h(z)}{(z-2)^2} dz = 2\pi i h'(2).$$

Now

$$\begin{aligned} h'(z) &= \frac{e^z}{(z-5)^5} - \frac{5e^z}{(z-5)^6} \\ &= \frac{e^z(z-10)}{(z-5)^6}. \end{aligned}$$

Thus

$$h'(2) = \frac{e^2(2-10)}{(2-5)^6} = -\frac{8e^2}{729}.$$

Consequently,

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz = -\frac{16\pi i e^2}{729}$$

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9. Compute the integral in the previous exercise with γ the closed path

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi.$$

Solution. The path

$$\gamma(t) = e^{it}, \quad 0 \leq t \leq 2\pi,$$

is the positively oriented unit circle $|z| = 1$. The integrand

$$\frac{e^z}{(z-2)^2(z-5)^5}$$

has singularities only at $z = 2$ and $z = 5$. Since

$$|2| > 1 \quad \text{and} \quad |5| > 1,$$

both singularities lie outside γ . Therefore the integrand is analytic on and inside the unit circle. By Cauchy's theorem,

$$\int_{\gamma} \frac{e^z}{(z-2)^2(z-5)^5} dz = 0$$

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